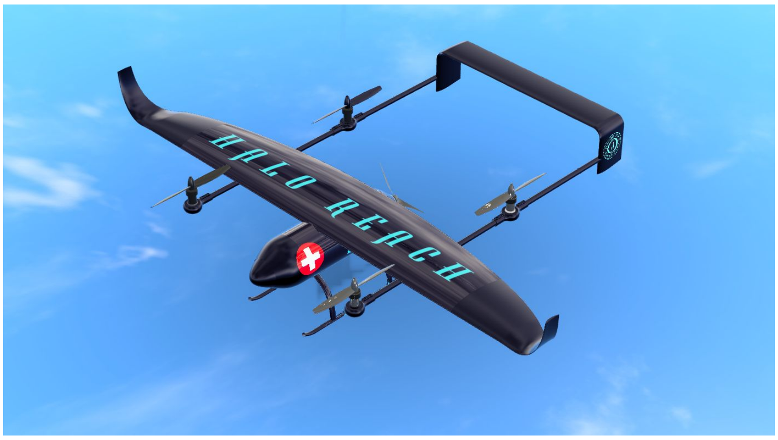
HALO REACH: V-n Diagram, Tail, Rotor and Propeller Loads and Structural Design

UAS for Disaster Relief - AE3 2024



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Abstract

This report presents the structural design and analysis of the tail, rotor and propeller for the HALO REACH Vertical Take-Off and Landing (VTOL) Unmanned Aerial System (UAS) developed for disaster relief operations. The design process involved creating a V-n diagram to delineate the aircraft's performance envelope, ensuring safe operation under various loading conditions. The rotor blades, designed as fully composite structures with an 8-layer symmetrical layup of 3960 prepreg composite, and the propeller blades, designed using a sandwich structure with a PVC core and the same composite skin, were subjected to rigorous load analysis. Additionally, the tail, designed using the NACA 0009 airfoil with a constant chord of 40 cm and a full span of 2.3 m, was analysed using similar methodologies. Finite Element Analysis (FEA) was employed to verify the structural soundness of the components, evaluating static loading and buckling. The results confirmed that the rotor blades, with a thickness of approximately 1 mm, and the propeller blades, with varying thickness from 15 mm to 1.12 mm, and the tail, meet the required performance criteria, providing robustness and reliability in disaster relief scenarios.

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1 Introduction

In the context of disaster relief operations, rapidly deploying supplies and establishing communications in areas with compromised infrastructure is crucial. The 2023 earthquakes in Turkey underscored the critical need for versatile aerial delivery systems capable of operating in challenging conditions where conventional transportation methods fail [1]. The Vertical Take-Off and Landing (VTOL) Unmanned Aerial System (UAS) designed for this project aims to fulfil these needs. This report focuses on the structural aspects of the aircraft, particularly the Velocity-load (V-n) diagram, tail, rotor and propeller.

Creating a V-n diagram was critical for understanding the aircraft's performance envelope and ensuring that it operates safely within its structural limits under various flight conditions. The primary goal following understanding the loads was to ensure the aircraft's structural integrity and performance under various loading conditions which was crucial for its safe operation in emergency scenarios. This was followed by component design which involved optimising the skin-stringer panels and ribs between strength and weight for tail. It also included rotor and propeller structural design. The structural elements and interfaces ensured that the aircraft can be built to precise specifications.

2 V-n Diagram

The V-n diagram in Figure 1 shows the limits of aircraft structural design in its mission profiles. V_A is the manoeuvring speed, V_B is the design speed for maximum gust intensity, V_C is the cruise speed, V_D is the design diving speed. No specific regulations on limit load factors was found for our aircraft, thus after discussion with the Flight Dynamics and Control Team, it was agreed that no drastic or acrobatic style of manoeuvre would be aimed during both the endurance and supply mission profile. Thus, FAR regulations were followed to have the limit load factors of 2.5 and -1 [2]. This diagram was only for cruise. During take-off and landing, the regulation on Helicopter Handbook was followed, which we considered a limit load of 3.5 and –0.5 [3].

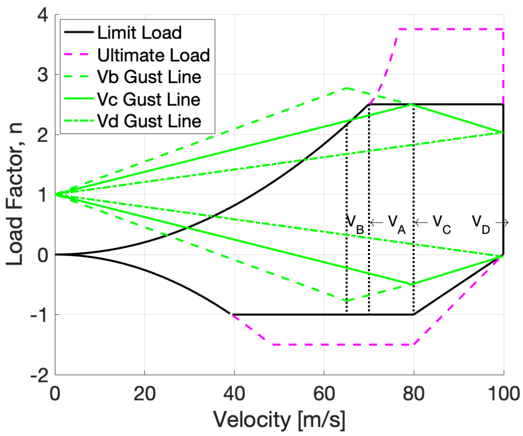

Figure 1. V-n diagram during cruise condition

3 Horizontal Tailplane

The geometry of the horizontal tailplane was designed by the Flight Dynamics and Control Team [4]. The twin boom tail utilised a NACA 0009 airfoil. The design featured a constant chord of 40 cm and a full span of 2.3 meters with a mass of 1.64 kg, resembling an airfoil shaped rectangular plate.

3.1 Horizontal Tailplane Loading

The main assumption of the horizontal tailplane was its idealisation as a double fixed end beam because the roots are connected to the vertical tailplanes, as shown in Figure 2. The idealised beam was subject to an evenly distributed inertial load and an elliptical aerodynamic load. Figure 3 presents a visualisation of the above loads. The inertial load rooted from the 5.94 kg mass of the horizontal tail and the aerodynamics load came from the 16 kg lift that was estimated by the Aerodynamics Team. In one engine inoperative (OEI) conditions, the horizontal tail experienced a 15 N side force exerted by the vertical tailplane estimated by the Aerodynamics Team.

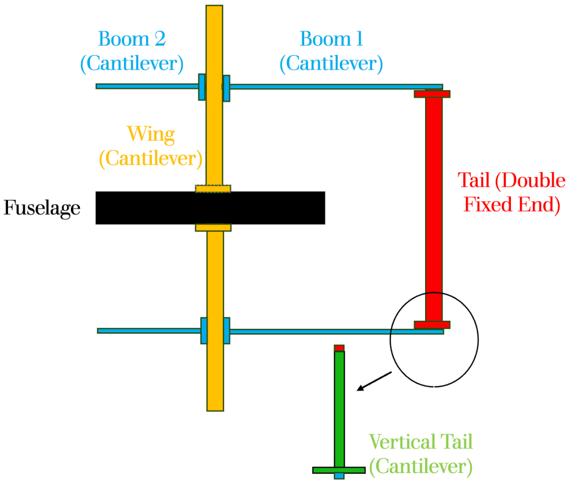

Figure 2. Idealisation of aircraft components for structural analysis

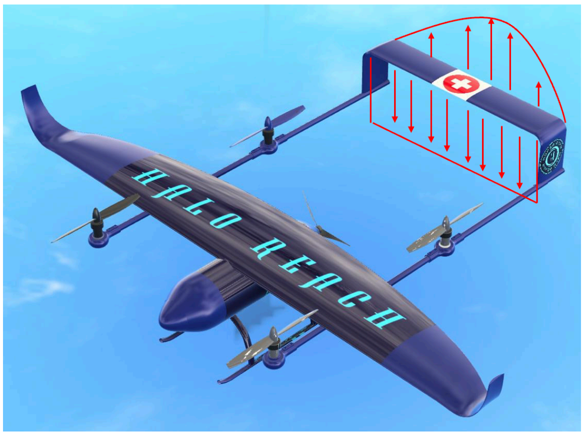

Figure 3. Types of load on the horizontal tailplane

Load conditions during take-off, cruise, and landing phases were considered with load factors of $n = 1$, -1.5 and 3.75. The take-off and landing phases considered six load cases which includes full fuel with load, full fuel with no payload, OEI of the front and rear rotor for each of two payload conditions. During cruise, for the supply mission, full fuel with full payload and zero fuel with zero payload was evaluated because these are the most constraining two cases. For the endurance mission, Maximum Take Off Weight (MTOW) and Maximum Zero Fuel Weight (MZFW) were considered.

The lift generated by the horizontal tailplane was modelled in Equation (1) where $L_0{}'$ was the sectional lift at the centre-span, given by Equation (2). The total tail lift was the integration of sectional lift - Equation (3). This was evaluated with substitutions for $L'(s)$ and $\Gamma(s)$ from Equations (4) and (5) respectively, and the result rearranged for Γ_0. Finally, it was clear that at $s = 0$, by Equation (4), $L_0{}' = \rho U_\infty \Gamma_0$, and substituting the previous result for Γ_0 yielded Equation (2) [5] [6].

$$L'(s) = L_0{}'\sqrt{1 - \left(\frac{s}{S_{half}}\right)^2} \quad (1) \qquad L_0{}' = \frac{2nL_w}{\pi S_{half}} \quad (2) \qquad L_w = \int_{-S_{half}}^{S_{half}} L'(s)ds \quad (3)$$

$$L'(s) = \rho V \Gamma(s) \quad (4) \qquad \Gamma(s) = \Gamma_0\sqrt{1 - \left(\frac{s}{S_{half}}\right)^2} \quad (5) \qquad L_w = \frac{1}{2}C_{Lw}\rho V^2 S_{ref} \quad (6)$$

The shear force (SF) and the bending moment (BM) experienced by the horizontal tailplane were then calculated, where BM was the integration of SF. Figure 4, 5 and 6 presents the BM in cruise, take-off and landing conditions. Figure 7 and 8 shows the max and min BM and SF during the three phases. It can be seen that the case with the highest BM, and thus the most constraining would be the n = 3.75 MTOW case during cruise. The torque, T, shown in Figure 9 was caused by the offset of the lift and inertial load from the flexural centre, which was assumed to be 50 % of the chord.

3.2 Horizontal Tailplane Component Design

Material Selection

3960 Prepreg System composite material was applied on all components of the horizontal tail given its advantages over other materials [7].

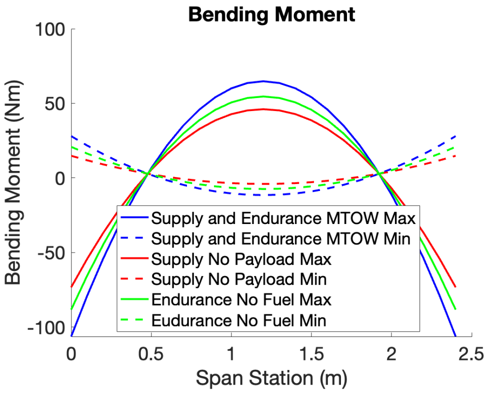

Figure 4. BM during cruise

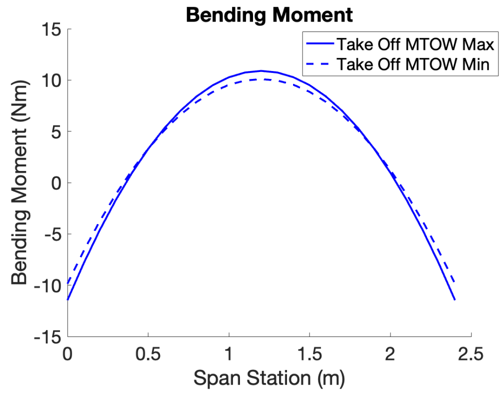

Figure 5. BM during take-off

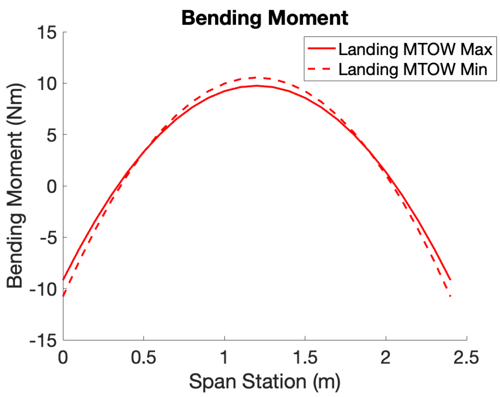

Figure 6. BM during landing

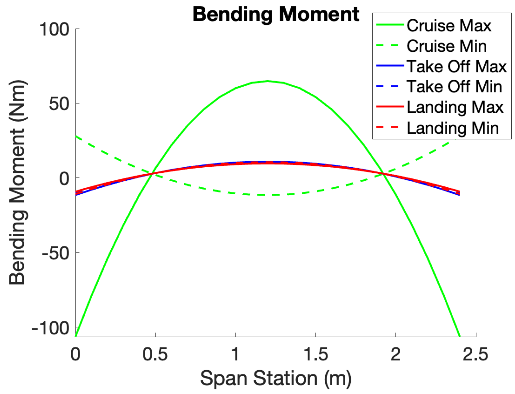

Figure 7. Max and min BM in cruise, take-off and landing

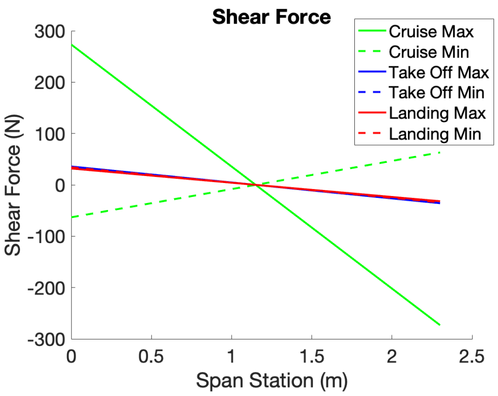

Figure 8. Max and min SF in cruise, take-off and landing

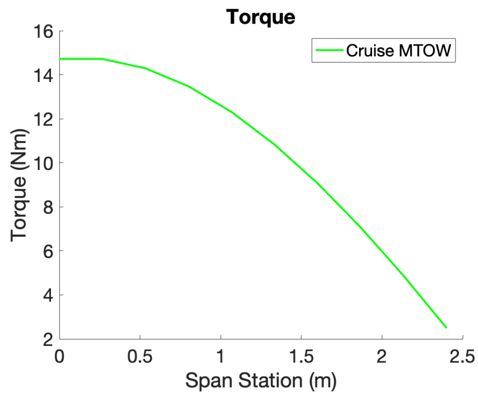

Figure 9. Torque in the most constraining case

Skin-Stringer Panel Design

To find an optimal geometry of the horizontal tailplane, a cost function was generated. The cost value, M_{cost} assessed the mass of the skin-stringer panels and ribs, as shown in Equation (7). The input of this function was an array of skin thicknesses because the optimisation calculations started from initialising a skin thickness and compute the corresponding stringer and ribs design from it.

$$M_{cost} = \int_0^{Half\ span} (A_{skin}\,\rho_{skin} + A_{stringer}\,\rho_{stringer})ds\; + \sum_{i=1}^{n_{ribs}} t_{rib}\,\rho_{rib}\,A_{rib} \tag{7}$$

The design of skin-stringer panels was a critical aspect, providing the necessary strength and stiffness to the tail structure. Initially, Z-type stringers were selected due to their optimal strength distribution, which contributed to a lighter yet robust structure [8] [9]. The skin thickness (t_s) was initially estimated and optimised based on the compressive load per unit width (N) and the bending moment (BM) experienced across the structure. The internal stress within the skin (σ_0) was calculated using Equation (8). The unit compressive load was estimated using the knowing bending moment across the span in Equation (9), where b_{bw} and b_{bh} were the penal width and height. This formed the basis for determining the stringer pitch (b_s) through Equation (10), where K_c was the compression buckling coefficient [8].

$$\sigma_0 = \frac{N}{t_s} \quad (8) \qquad N = \frac{BM}{b_{bw} \times b_{bh}} \quad (9) \qquad b_s = \sqrt{\frac{K_c E {t_s}^2}{\sigma_0}} \quad (10)$$

The dimensions of the stringers were then calculated, with the stiffening ratio (SR) which was the ratio of the stiffener area (A_{st}) to the skin area (A_s). The thickness of the stringer flanges were determined using Equation (11), and the flange widths, b_f, were defined by Equation (12) [10]. These calculations ensured the skin-stringer panels meet the structural requirements while optimising material usage.

$$t_f = 0.7 t_s \quad (11) \qquad b_f = b_a = \begin{cases} 2.08 t_s + 0.0175, & \text{if } t_s \leq 0.0076 \\ 0.333, & \text{if } t_s > 0.0076 \end{cases} \quad (12)$$

The stringer web dimensions depended on the effective skin width, b_e which could be calculated using Equation (13). η is the plasticity correction factor and is defined as the ratio between the tangential modulus, E_t and Young's modulus, E of the material [10]. The stringer web length, b_w was calculated using Equation (14), and the stringer web thickness, t_w was calculated using Equation (15) [10].

$$b_e = t_s \sqrt{\frac{K \eta E}{\sigma_{cr}}} \quad (13) \qquad b_w = \sqrt{\frac{b_e}{t_s} \left(A_{st} - 1.4 b_a t_s\right)} \quad (14) \qquad t_w = b_w \left(\frac{t_s}{b_e}\right) \quad (15)$$

Figure 10 shows the first and primary optimisation outcome where the minimum mass can be reached with a skin thickness of 1.5 mm. The red dotted line at 1 mm represents the engineering constraint. Given the skin thickness, the stringers pitch was then found using the method explained above, shown in Figure 11. The minimum pitch 0.06 m was adopted as a constant pitch across the tail. The stringer flange thickness needed ideally was less than 1 mm. However, due to the engineering constraints, 1 mm was determined to be the thickness, as seen in Figure 12. The skin-stringer panel design was summarised in Table 2 and a schematic of the horizontal tailplane can be seen in Figure 13.

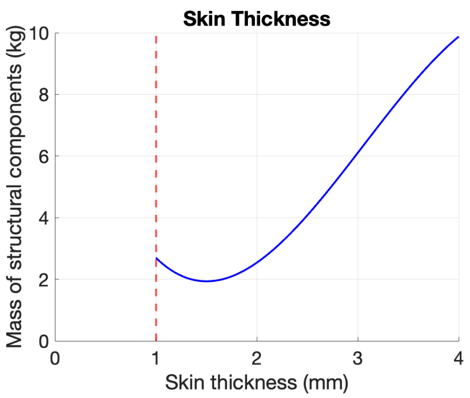

Figure 10. Total tail mass with different skin thicknesses

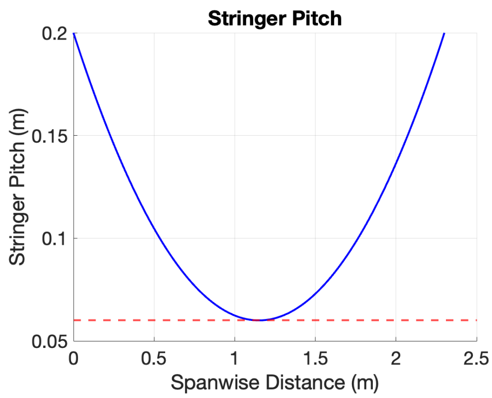

Figure 11. Minimum stringer pitch required across span

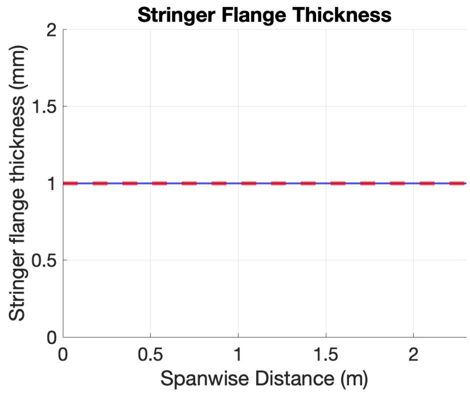

Figure 12. Flange thickness across span

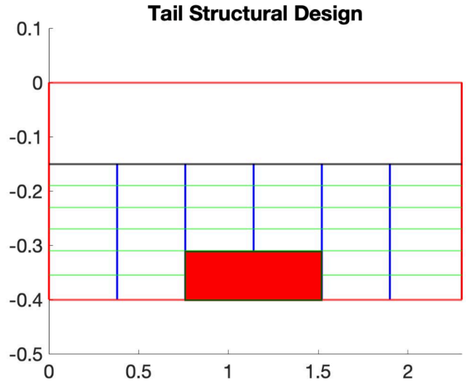

Figure 13. Schematic of the horizontal tailplane

Table 2. Skin-stringer Design

	No. of layers	Thickness	Layup
Skin	8	1.5 mm	$[45\ -45\ 90\ 0]_s$
Stringer flange	6	1.1 mm	$[0\ 90\ 0]_s$
Stringer web	8	1.5 mm	$[45\ -45\ 90\ 0]_s$
Stringer pitch	60 mm		
Stringer width	2 mm		

Rib Design

Ribs played a crucial role in providing support and stability to the skin-stringer panels. The design process involved calculating the crushing force (F_{crush}) using Equation (16).

$$F_{crush} = \frac{BM^2 L b_{bw} b_{bh} T}{2EI_{st}} \tag{16}$$

where $T = t_s + \frac{A_{st}}{b_s}$. This force was critical in determining the rib thickness (t_{rib}), derived from the Equation (17). The rib spacing, L resulted from equating structural stability to the flexural buckling stress, σ_f from the Farrar efficiency Equation (18) to the applied stress at failure [8].

$$t_{rib} = \left(\frac{F_{crush} {b_{bh}}^2}{KEb_{bw}}\right)^{\frac{1}{3}} \tag{17}$$

$$F = \sigma_f \sqrt{\frac{L}{NE_t}} \tag{18}$$

The ribs optimisation yielded results that had to align with the engineering constraint, similar to Figure 12. The ribs had a lay up of $[0\ 90\ 0]_s$ giving 1.1 mm of thickness and with a spacing of 0.42 m.

Spar Design

The spar design was essential for ensuring the tail withstand bending and torsional loads. A single spar was placed 20% of the chord to maximise torsional stiffness. The height of the spars were constrained by the thickness of the horizontal tailplane which was 3.6 cm. The single spar required the ribs to be riveted on to be spar to reduce torsional displacement.

The web and flange dimensions were again determined by using the unit BM and SF experienced in discrete panels and to prevent global buckling happening. These dimensions were crucial in accommodating the aerodynamic loads during flight. The spar featured 8-layer symmetric layup of $[45\ -45\ 90\ 0]_s$ giving 1.5 mm of thickness for both the spar web and flanges of width 2 mm.

Design Validation

To ensure the overall skin-stringer panel and rib design were sufficient for the loads applied, a stress ratio criteria needed to be met. The compressive, shear, and bending stresses - σ_c, σ_s, and σ_b must not exceed their relative critical stresses, but there was also a combined effect of all three acting on the same component as shown in Equation (19) [10]. The final optimised components presented above yielded a stress ratio of 0.91, which fulfilled the safety requirement for the structure.

$$\frac{\sigma_s}{\sigma_{s,cr}}^2 + \left(\frac{\sigma_c}{\sigma_{c,cr}} + \frac{\sigma_b}{\sigma_{b,cr}}\right)^2 \leq 0.99 \tag{19}$$

where

$$\sigma_c = \sigma_0 \quad (20) \qquad \sigma_s = \frac{SF}{A_{st} + b_s t_s} \quad (21) \qquad \sigma_b = \frac{BM}{A_{st} + b_s t_s} \quad (22) \qquad \sigma_{i,cr} = K_i E \left(\frac{t_s}{b_s}\right)^2 \quad (23)$$

3.3 Computational Validation

Finite Element Analysis (FEA) was also conducted on Abaqus to validate the design. This analysis included static loading tests and buckling evaluations.

The static loading tests were performed to verify that the rotor and propeller blades can withstand the operational loads without excessive deformation or failure. The buckling analysis was carried out to assess the stability of the structures under compressive loads, ensuring that the blades will not experience sudden catastrophic failure.

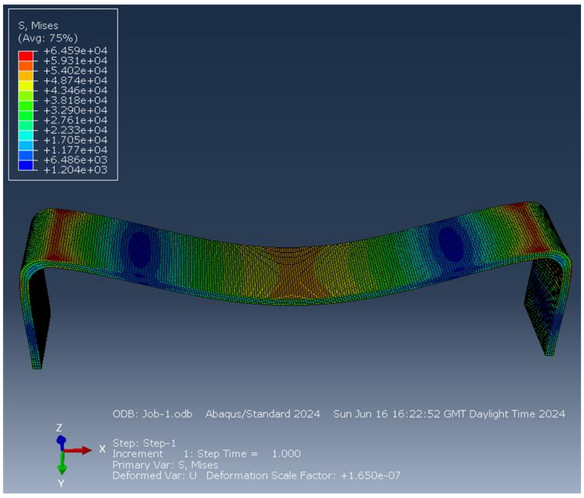

Figure 14. Stress of the tail

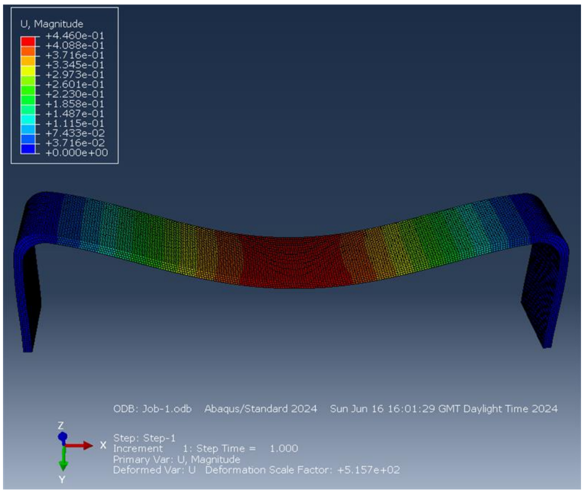

Figure 15. Displacement of the tail

The maximum stress and displacement the tail experiences are 64590 Pa and 0.446 m respectively, as evident in Figure 14 and 15. The critical buckling load is 3622 N, taken from the eigenvalues in the simulation, as seen in Appendix A. By conducting these analyses, we have verified that the designs are structurally ensuring both safety and performance, and it was ensured that the global mesh size of 1.1 utilised in FEA was converged.

4 Vertical Tailplane

The geometry of the vertical tailplane was designed by the Flight dynamics and Control Team [4]. NACA 0009 airfoil was used with a constant chord of 40 cm and a height of 55 cm. The load on the vertical tailplane was solely transmitted from the horizontal tailplane. Therefore, the vertical tailplane was idealised as a fixed end cantilever beam, as seen in Figure 2. The loads it experienced included the compression reaction force and the bending moment transmitted from the horizontal tailplane, and the aerodynamic force generated by itself in OEI cases. Figure 16, 17, 18 presents the loading on the vertical tailplane in the most constraining cases.

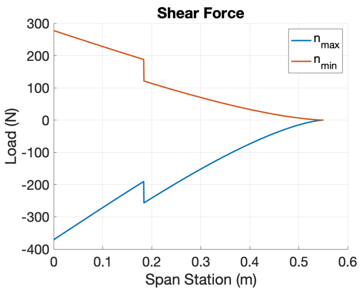

Figure 16. Shear Force

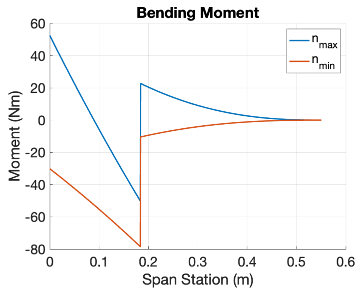

Figure 17. Bending Moment

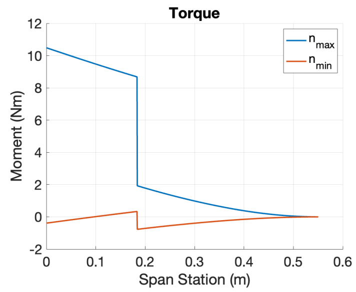

Figure 18. Torque

The design of the spars, skin-stringer panels and the ribs utilised the same optimisation method as explained in the Horizontal Tailplane Section. Table 3 presents the design of a 1.04 kg vertical tailplane configurations to ensure its structural integrity.

Table 3. Components of the vertical tailplane

	No. of layers	**Thickness**	**Layup**
Skin	8	1.5 mm	$[45\ -45\ 90\ 0]_s$
Stringer flange	6	1.1 mm	$[0\ 90\ 0]_s$
Stringer web	8	1.5 mm	$[45\ -45\ 90\ 0]_s$
Stringer pitch	60 mm		
Stringer width	2 mm		
Ribs	6	1.1 mm	$[0\ 90\ 0]_s$
Rib spacing	0.2 m		
Spar web	8	1.5 mm	$[45\ -45\ 90\ 0]_s$
Spar flange	8	1.5 mm	$[45\ -45\ 90\ 0]_s$

5 Rotor and Propeller

The aircraft features four rotors for take-off and landing and a pusher propeller for cruise. Their geometries were designed by the Aerodynamics Team [11]. The rotor has two blades, with a blade span of 0.42 m and incorporating a Clark Y airfoil being transited to NACA 23012 along the span with the blade's twist angle varying from 49.12° to 19.14° from the root to the tip with a thickness varying from 1.2 mm to 1.05 mm. The propeller has three blades with a blade span of 0.44 m and incorporated an E216 airfoil. The rotor blade's twist angle varied from 79.35° to 34.38° from the root to the tip with a thickness varying from 15 mm to 1.12 mm. The propellers and rotors are shown in Figure 19 and 20.

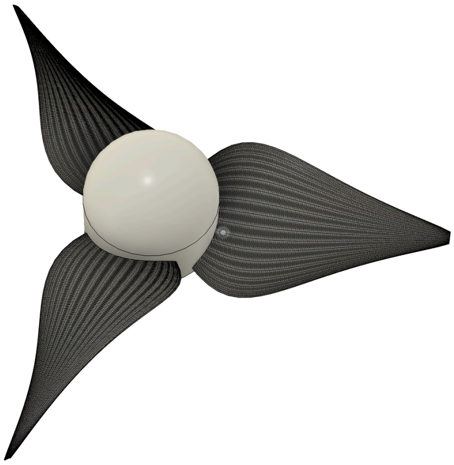

Figure 19. Propeller CAD

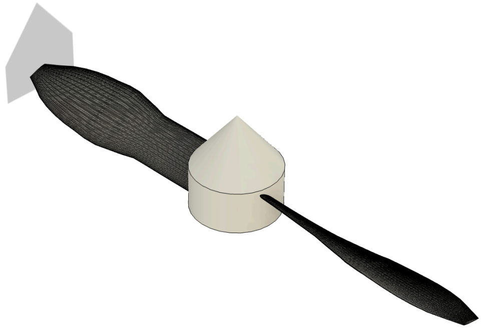

Figure 20. Rotor CAD

5.1 Rotor and Propeller Loading

Designing the rotor blade for VTOL and the pusher propeller for forward flight involved understanding and addressing various load conditions. This section outlines the different types of load conditions experienced by rotor blades and propellers.

Flapwise Loads

Flapwise loads were primarily due to aerodynamic forces acting perpendicular to the rotor blade's plane of rotation. Critical load factors ranged from -0.5 to 3.5 for take-off and landing case and -1.5 to 2.5 for cruise, with the total flapwise load being a product of these factors and lift. BM, SF and Torque acting on the blades were then computed and shown in Figure 21 based on the lift distribution given by the Aerodynamics Team [11].

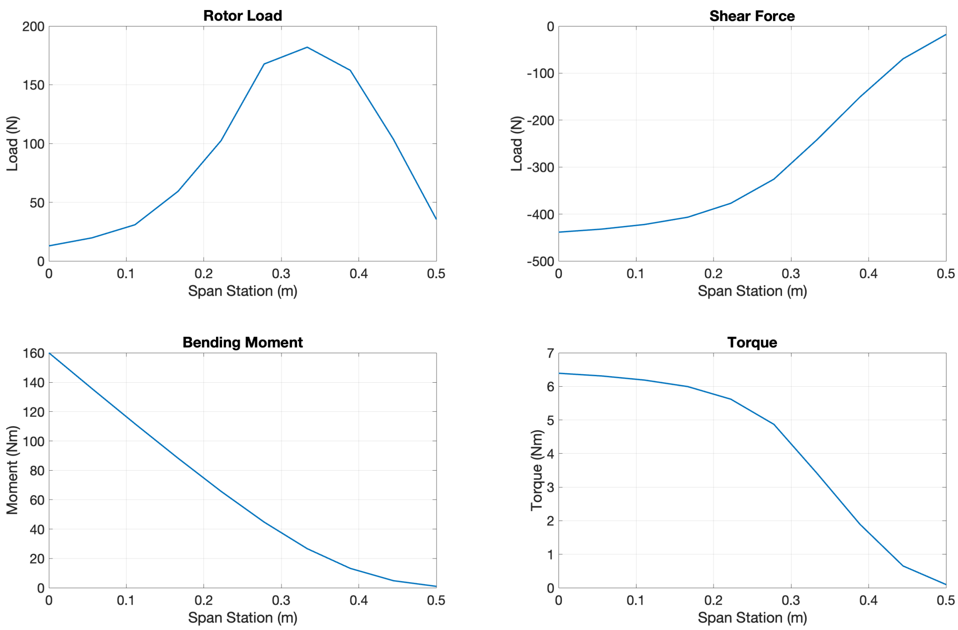

Figure 21. Flapwise load of the rotor

In-Plane Loads

In-plane loads was caused by the transmission of shaft torque from the power plant to the rotor blades. There were two primary scenarios: one involving rotor acceleration due to increased power, creating an in-plane moment at the blade root, and the other involving braking torque being transmitted equally to all blades. These loads resulted in inertial forces distributed along the blade span, leading to in-plane inertial loads, shown in Figure 22, that must be balanced against the generated root moments. The limit root in-plane moment M_E was

$$M_E = \frac{1.5 M_T}{b-1} \tag{24}$$

where M_T was the the torque at military power rating of the power plant. The angular acceleration $\dot{\Omega}$ was then decided by Equation (25), and thus the in-plane load could be found by Equation (26) [12].

$$M_E = \sum_{i=1}^{n} m_i r_i^2 \dot{\Omega} \tag{25}$$

$$q_i(r) = \frac{m_i r_i \dot{\Omega}}{l_i} \tag{26}$$

Torsional Loads

Torsional loads were generated by two main factors: aerodynamic pitching moments and rotor moments due to centrifugal forces. Rotor moments resulted from the distribution of the blade mass both forward and aft of the elastic axis, causing centrifugal forces that create torsional stresses. These forces attempt to flatten the blade back into its rotational plane, a phenomenon known as centrifugal flattening.[12] The torsional load on the rotor is shown in Figure 23. and Equation (27) shows the calculation of the torsional load.

$$\mathcal{T}_{PM_i} = I_{\theta_i} \Omega^2 \theta_i \tag{27}$$

Centrifugal Loads

Centrifugal forces acted axially along the blade when it rotated, creating significant tensile stresses. These loads were combined with flapwise, in-plane, and torsional load cases rather than being considered separately. The magnitude and distribution of centrifugal loads were determined based on the maximum rotor rotational velocity and the blade's deformation due to other loads. The centrifugal load of the rotor is presented in Figure 24. The magnitude of the centrifugal load was governed by Equation (28)

$$\mathrm{CF}_i = m_i r_i \Omega^2 \tag{28}$$

where i is the segment of the blade[12].

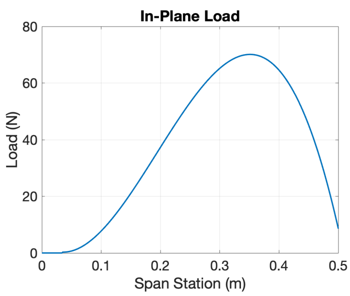

Figure 22. Inplane laod

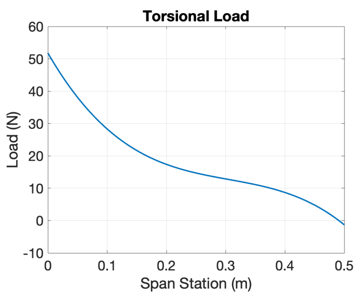

Figure 23. Torsional load

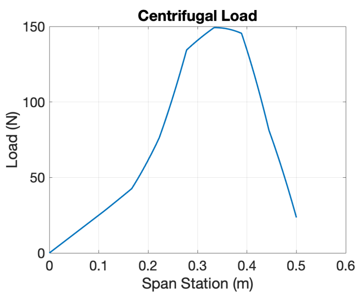

Figure 24. Centrifugal load

Non-Flight Loads

Non-flight loads accounted for adverse conditions such as ground handling, stop-banging, turning the rotor at low speeds in strong winds, etc. Design required the rotor blade to withstand static loads equal to its weight multiplied by a limit load factor of 4.67 [12]. The constraining non-flight load is 32.07 N.

5.2 Rotor and Propeller Design

The rotor was fully composite due to the engineering constraint of a maximum thickness of 1 mm, which precluded the use of a sandwich core. Conversely, the propellers, with a thickness ranging from 15 mm to 1.12 mm, was designed using a sandwich structure with a PVC core. The skin for both rotors and propellers used the 3960 Prepreg composite system [7]. This section outlines the detailed design considerations and specifications for the rotor and propeller structures.

Rotor Design

The rotor blade design addressed the various loads explored in the above section: flapwise, in-plane, torsional, centrifugal, and non-flight loads.

Composite Layup for Rotor Blades:

- **Material:** 3960 Prepreg system composite
- **Layup Schedule:** A symmetric layup pattern was chosen for balance and structural integrity.
  - **Layer Sequence:** [45°, -45°, 0°, 90°, 90°, 0°, -45°, 45°]
  - **Total Layers:** 8 layers

This symmetric layup ensured balanced properties and helped mitigate warping or twisting under load. The orientation of fibers at 45° and -45° angles provided resistance to shear loads, while 0° and 90° layers offered strength in axial and transverse directions, respectively.

Propeller Design

For the propellers, the varying thickness from 15 mm to 1.12 mm allowed the use of a sandwich structure with a PVC core to improve stiffness and reduce weight. The skin was the same 3960 Prepreg system composite used for the rotor blades.

Composite Layup for Propeller Blades:

- **Layup Schedule:** Similar to the rotor blades, a symmetric layup is used.
  - **Layer Sequence:** [45°, -45°, 0°, 90°, 90°, 0°, -45°, 45°]
- **Core Thickness:** For the maximum blade thickness (15 mm), the core will be approximately 13 mm thick, leaving 1 mm thick skins on each side. For the minimum blade thickness (1.12 mm), a solid composite section without a core was used. The thickness of the core will taper smoothly from the root to the tip to match the thickness profile of the propeller blade.

5.3 Design Validation and computational results

Similar FEA tests as in Section 3.3 using a converged mesh size of 11 were conducted on the rotor and propeller design. Results for rotor and propeller are presented in Figure 25, 26, 27, and 28, with maximum displacements of 4.23 cm and 10.64 cm and critical buckling load of 2423.6 N and 1036.6 N respectively. Stress on the rotor and propeller were relatively small, as seen in Appendix A. The design of the rotor and the propeller was thus verified to be structurally strong enough.

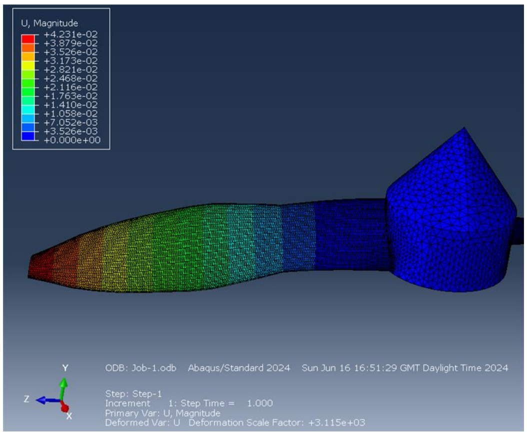

Figure 25. Displacement of rotor

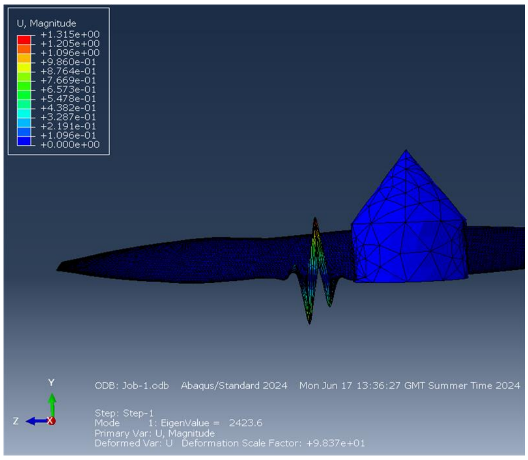

Figure 26. Critical buckling load of rotor

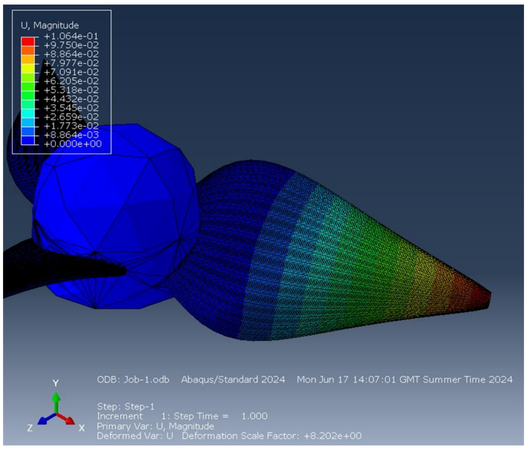

Figure 27. Displacement of propeller

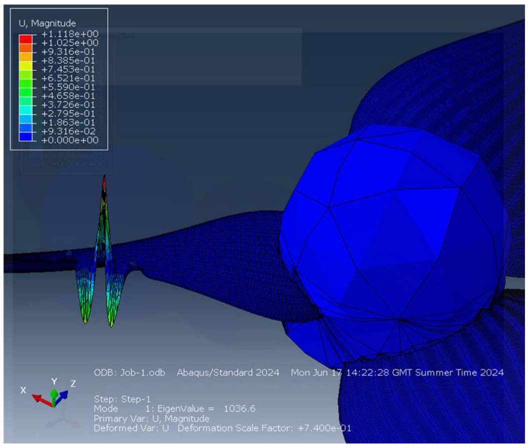

Figure 28. Critical buckling load of propeller

6 Improvement of the Overall Design

6.1 Tailplane Design

In the tailplane loading calculations, the shear center was assumed to be at the center, which could be recalculated in the next design iteration depending on the present spar thicknesses. The stringer design could be further improved by considering varying stringer pitch rather than using the minimum spacing required. This approach will provide more flexibility in the design of the stringers and result in a lower-weight design. Additionally, employing Z-type and J-type stringers on the upper and lower surfaces to better suit the particular stress needs of those surfaces will reduce the weight of the stringers but produce a more complex design. Although the rib design met the weight requirement, cutouts spaced along the ribs to enhance the buckling strength can be considered, although stresses tend to concentrate around these cutouts. The current iteration is acceptable since the critical shear buckling stress τ_{cr} was less than the material's Tresca shear yielding stress τ_{Tresca} for any realistic number of pseudo-ribs and at any spanwise station in the tailplane. However, new skin thickness distributions would ideally be iterated over, such that τ_{cr} and τ_{Tresca} equate.

6.2 Rotor and Propeller Design

While the current design of the rotor and propeller are robust, there are many potential issues with composite structures and sandwich construction to concern.

Delamination: Delamination is a critical issue in composite structures, particularly under cyclic loading conditions. To mitigate this, future designs could incorporate toughened resin systems or interleaved layers to improve the interlaminar shear strength. Additionally, non-destructive testing methods such as ultrasonic inspection should be routinely employed to detect early signs of delamination.

Thermal Stresses: Thermal stresses can arise due to the different coefficients of thermal expansion (CTE) between the composite skins and the PVC core in sandwich structures. These stresses can lead to issues such as skin-core debonding, shown in Figure 29. To address this, the selection of materials with more compatible CTEs or the inclusion of thermal barriers can be considered. Furthermore, thermal cycling tests should be conducted to assess the long-term performance of the sandwich structure under operational temperature variations.

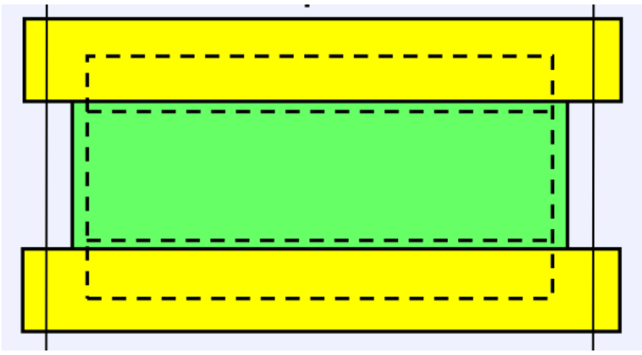

Figure 29. The skin and the core debond under thermal expansion

Manufacturing Considerations: The manufacturing process for the rotor and propeller blades should ensure consistent quality and minimise defects. Automated fiber placement (AFP) or automated tape laying (ATL) technologies should be reseached and explored to enhance the precision and repeatability of the composite layup process [13]. Additionally, ensuring proper curing cycles and environmental controls during manufacturing can prevent defects related to resin curing and fiber misalignment.

Optimisation of Core and Skin Thickness: The current design assumes uniform skin thickness, but future iterations could explore optimising the thickness of the skins and core based on stress analysis results.

Integration with Actuation Systems: For the propeller, the integration of actuation systems, such as pitch control mechanisms, should be considered. This involves ensuring that the design accommodates the mechanical interfaces and load paths associated with these systems.

Environmental and Operational Durability: The design should also account for environmental factors such as moisture absorption and UV exposure, which can degrade composite materials over time. Protective coatings and sealants can be applied to enhance durability. Additionally, operational factors such as impact resistance and abrasion should be considered, particularly for components exposed to debris and other foreign objects during the flight of a disaster relief mission.

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Appendix

A Finite Element Analysis Results of Tail, Rotor and Propeller

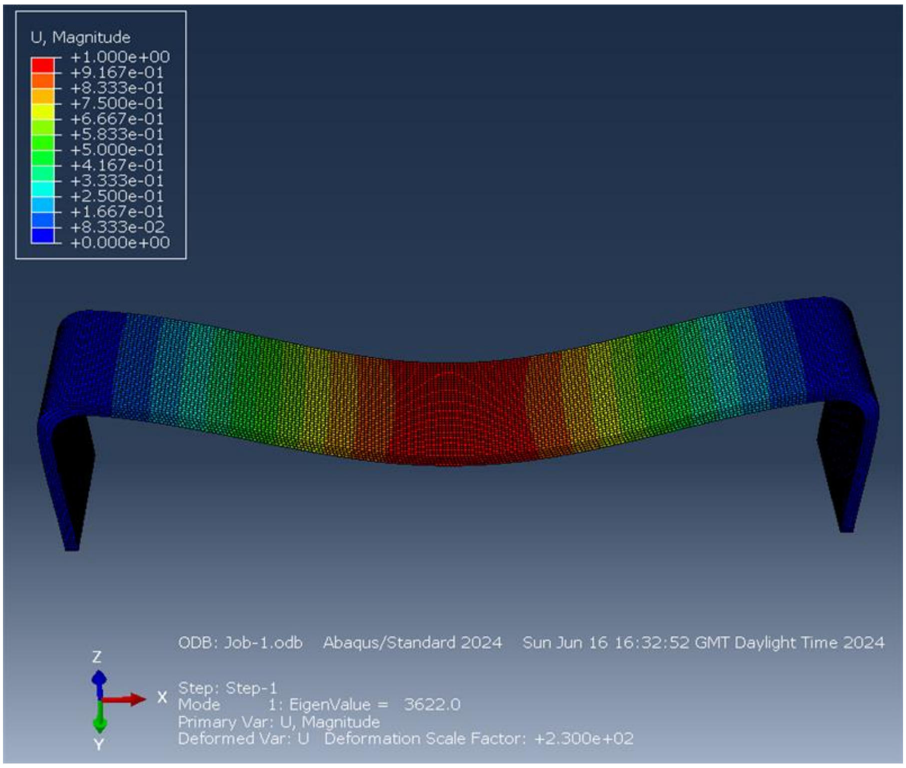

Figure 30. Buckling test for tail

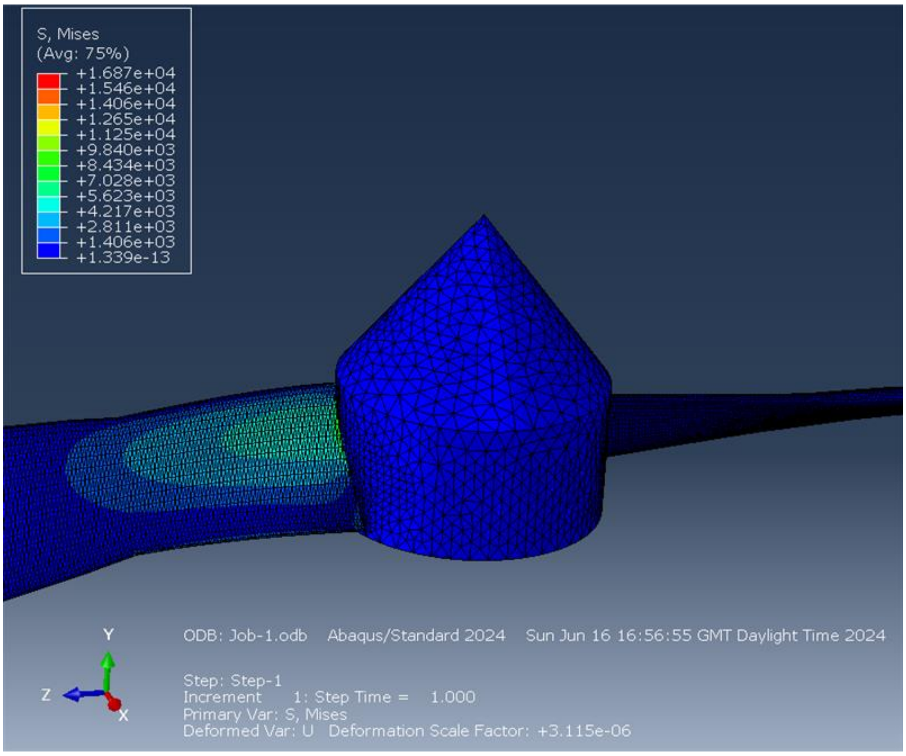

Figure 31. Stress Distribution on rotor

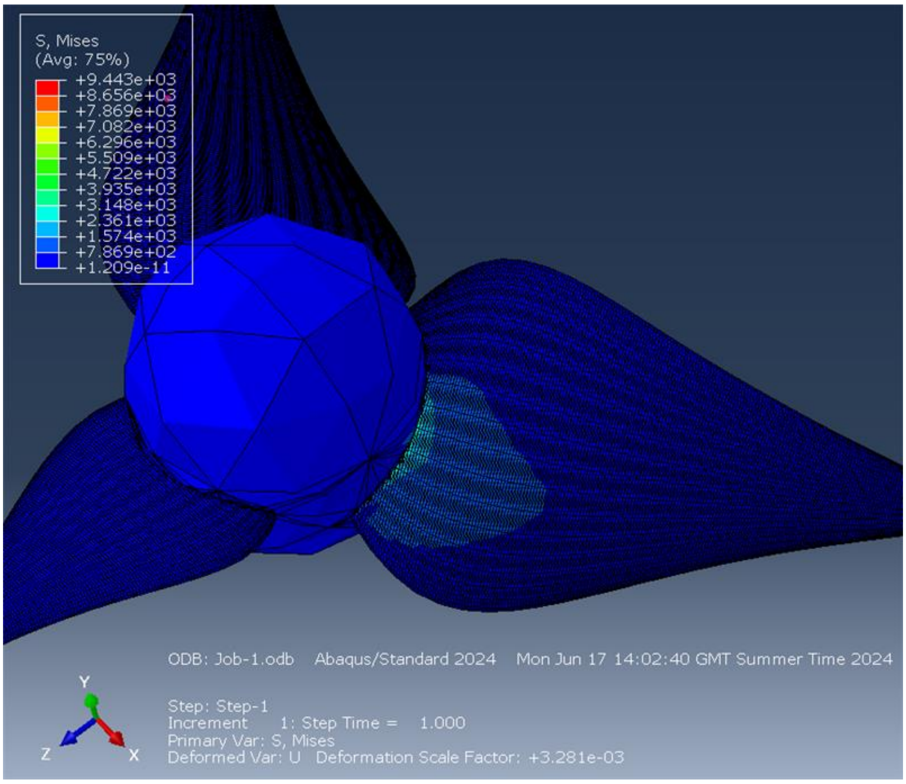

Figure 32. Stress Distribution on propeller